ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Alliance School of Advanced Computing

Capstone Project – I 7CS1999

Project Title: An AI-Powered Crop Disease Detection and Agrochemical Recommendation Framework

Batch : 2023 - 2027

Batch No : ASAC-CP-009

Presented by

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I. Project Overview

- **Problem Context** – Farmers may struggle to identify crop diseases early and select the appropriate agrochemical treatment. Traditional diagnosis can be time-consuming and dependent on expert knowledge.
- **Motivation** – To develop an AI-based system that can detect crop diseases from plant images and provide suitable agrochemical recommendations.
- **Significance** – Helps in early disease detection, reduces unnecessary agrochemical usage, supports better farming decisions, and protects agricultural data through secure communication.
- **Application Domain** – Smart Agriculture / Precision Agriculture / AI in Agriculture / Cloud Security
- **Target Users** – Farmers, agricultural consultants, agronomists, agricultural researchers, and farm-management organizations.

II. Objectives

Crop Disease Detection: Develop an AI-based system to identify and classify crop diseases from leaf images.

Image Preprocessing: Improve image quality and prepare the dataset for effective disease classification.

Smart Agrochemical Recommendation: Recommend suitable agrochemical treatments based on the detected crop disease using reliable agricultural information.

System Integration and Validation: Integrate disease detection and treatment recommendation modules and evaluate the system’s performance.

III. Literature Review

Author Name, Journal/Conference, Year	Research / Methodology	Dataset Used	Performance Metrics	Advantages	Disadvantages
1. S. K. Mahmudul Hassan and Arnab Kumar Maji, IEEE Access, 2022.	Novel CNN combining Inception layers, residual connections and depthwise-separable convolutions to reduce model parameters.	PlantVillage (corn, potato and tomato disease classes); rice disease dataset (5,932 images); cassava disease dataset (5,656 images).	Reported validation accuracy: 99.39% on PlantVillage, 99.66% on rice and 76.59% on cassava. The paper also evaluates precision, recall and F1-score.	Lightweight architecture with fewer parameters; tested across multiple crops; evaluated on both controlled and field-condition images.	Substantial performance drop on cassava; class imbalance and complex backgrounds affect generalization.
2. C. J. Xan et al., IEEE Access, 2024.	Compares MobileNet V1/V2, FD-MobileNet, ResNet and SqueezeNet, optimized and quantized for ARM-M microcontroller deployment.	5,932 rice-leaf images covering bacterial blight, blast, brown spot and tungro.	98.44% accuracy for FD-MobileNet.	Lightweight inference; designed for resource-constrained hardware and real-time, on-device disease detection.	Limited to rice diseases; hardware and memory constraints restrict the model choices and deployment scope.
3. R. Maurya, L. Rajput and S. Mahapatra, IEEE Access, 2025	RAI-Net: fine-tuned ResNet18 with channel attention and Inception modules; Grad-CAM provides visual explanations of predictions.	22,930 tomato-leaf images, 10 classes; test set of 4,595 images.	97.88% accuracy on the 4,595-image test set.	Multi-scale feature extraction; visual explanations help identify image regions influencing predictions; specialized for tomato diseases.	Limited to tomato in the reported experiment; broader validation under uncontrolled field conditions is still needed.

4. W. Xiao et al., IEEE Access, 2025.	**framework using class-agnostic contrastive supervised and sparse orthogonal mapping to reduce background interference and improve feature separation.**	**Newly introduced citrus-leaf dataset: 5,763 images across 6 categories, including field variations in illumination, occlusion and lesion scale.**	**97.1% accuracy, reported as a 2.4-percentage-point improvement over ResNet-18.**	**Specifically addresses complex field images; supports lesion localization and better separation of disease features.**	**Focused on citrus; additional large-scale testing across crops, regions and growing conditions is needed.**
5. T.-C. Pham et al., IEEE Access, 2025.	Iterative active learning with diverse expansion sampling to select informative, representative samples for annotation.	VIECOLD and BRACOL coffee-leaf disease datasets from different countries	F1-score improvement of 1.7% on VIECOLD and 0.9% on BRACOL using the same labeled-data BRACOL labels surpassed the corresponding full-data baseline.	Reduces manual labeling requirements; improves data efficiency and can lower annotation costs.	Requires repeated sample selection and human annotation; effectiveness depends on sample diversity and the baseline model.
6. Arfat Ahmad Khan, Muhammad Faheem, Rab Nawaz Bashir, Chitapong Wechtaisong and Muhammad Zahid Abbas, IEEE Access, 2022.	IoT-assisted, context-aware fertilizer recommendation. Compares Logistic Regression, SVM, Gaussian Naïve Bayes (GNB) and KNN using soil fertility, crop and soil-type information.	Soil fertility mapping and agricultural data associated with The Islamia University of Bahawalpur, including N, P and K observations, crop and soil type.	GNB achieved 96% training accuracy and 94% testing accuracy. Mean differences for N, P and K observations were 0.34, 0.36 and −0.13, respectively.	Incorporates soil context into fertilizer recommendations; compares several ML classifiers; supports more context-aware fertilizer management.	Depends on reliable soil measurements and regional data; recommendations may need recalibration for other locations and crops.

7. Ayesha Khaliq, Atif Khan, Salman Jan, Muhammad Umair, Asad Gulshair, Ahmad Ali and Usman Ali Shah, IEEE Access, 2025.	AI + IoT + explainable AI. Uses TabNet for fertilizer recommendation and soil-health analysis, SwiFT for crop recommendation, and a TTL model for irrigation advice.	Integrated agricultural data comprising temperature, humidity, soil moisture, soil type, crop type, N/P/K and fertilizer information.	TabNet fertilizer recommendation: 99.3% accuracy; SwiFT crop recommendation: 98.75% accuracy; TabNet soil analysis: R^2 = 98.7%; TTL irrigation: 99.13% accuracy	Integrates fertilizer, crop, irrigation and soil analysis; incorporates explainability; provides a broad decision-support framework.	Results depend on the training data and fertilizer labels; the reported metrics do not by themselves establish field performance across different regions.
8. Purnima Awasthi et al., IEEE Access, 2025.	Extended Multiscale Residual Convoluted Recurrent GAN for corn-yield forecasting, combined with a Hierarchical Fuzzy Model for N/P/K fertilizer recommendations.	ICRISAT agricultural data, NASA POWER weather data and Fertilizer Association of India data; the literature review reports 3,960 records.	The literature review reports approximately 12% improvement in prediction accuracy and 15% improvement in efficiency over traditional approaches.	Combines yield prediction with fertilizer optimization; integrates weather, crop and nutrient information; uses fuzzy rules for fertilizer decisions	Focuses on corn and Indian agricultural conditions; the reported improvements should be interpreted within the study's experimental setting.
9. Weixi Ge, Jun Zhou, Pengyuan Zheng, Agriculture, 2024.	case-based reasoning (CBR), Retrieves a preliminary fertilization scheme from the knowledge graph and refines it using similar historical cases.	Rice-fertilization knowledge and historical fertilization cases. The published abstract does not specify a named public benchmark dataset.	Fertilization-scheme error rate: approximately 19.5%. Prediction accuracy for fertilizer rates: N 95.69%, P_2O_5 79.04%, K_2O 81.67%.	Directly recommends fertilization plans agricultural knowledge; interpretable and designed to work with limited real-time field data.	Focused on rice fertilization; performance differs among nutrients, with lower accuracy for P_2O_5 and K_2O than for N.

IV. Research Gap

Identified Research Gap – Existing studies mainly focus on disease detection**,** while disease detection, agrochemical recommendation, are rarely integrated into one system.

Technical Limitations – Many existing models are trained on controlled datasets and may have reduced performance on real-field images. Recommendation systems may also depend on limited soil/crop information.

Need for Proposed Work – A system is needed that can provide early disease detection, suitable agrochemical recommendations, and secure transmission of agricultural data.

Proposed Differentiation – Our system combines AI-based disease detection + smart agrochemical recommendation in a single platform.

V. Research Gap Matrix

Research Area	Existing Research	Identified Research Gap	Proposed Solution
Crop Disease Detection	CNN and deep learning models classify crop diseases from leaf images.	Performance may decrease with complex backgrounds and real-field images.	Develop a multi-class disease classification system covering multiple crops.
Disease Classification	Existing models focus on specific crops or disease categories.	Limited generalization across different crops and disease classes.	Train and evaluate the model using a multi-crop dataset.
Agrochemical Recommendation	ML models recommend fertilizers using soil nutrients and crop information.	Disease-specific agrochemical recommendations are often not integrated with disease detection.	Recommend suitable treatments based on the detected crop disease.
System Integration	Disease detection and agricultural recommendation are often developed as separate modules.	Lack of a unified system connecting disease identification with treatment recommendations.	Integrate both modules into a single AI-powered application.
Practical Applicability	Many systems are evaluated using controlled datasets.	Limited consideration of field conditions, disease severity, and recommendation reliability.	Design an extensible system with scope for field-image validation and agricultural-context integration.

VI. Scope

- **Crop Disease Detection:** To develop a deep learning-based system that identifies crop diseases by analyzing uploaded leaf images.
- **Multi-Class Classification:** To classify crop leaf images into healthy and diseased categories across multiple crops using the EfficientNet-B0 model, supporting 38 predefined classes.
- **Smart Agrochemical Recommendation:** To provide disease-specific agrochemical suggestions and relevant management practices, including chemical, biological, cultural, and preventive measures, using a structured agricultural knowledge base.
- **Integrated AI-Based Application:** To integrate crop disease detection, recommendation generation, and Grad-CAM-based visual explanation into a single user-friendly application.
- **Future Enhancement:** To establish a foundation for incorporating soil properties, weather conditions, and disease severity in future versions to improve recommendation relevance and real-world applicability.

VII. Requirements Analysis

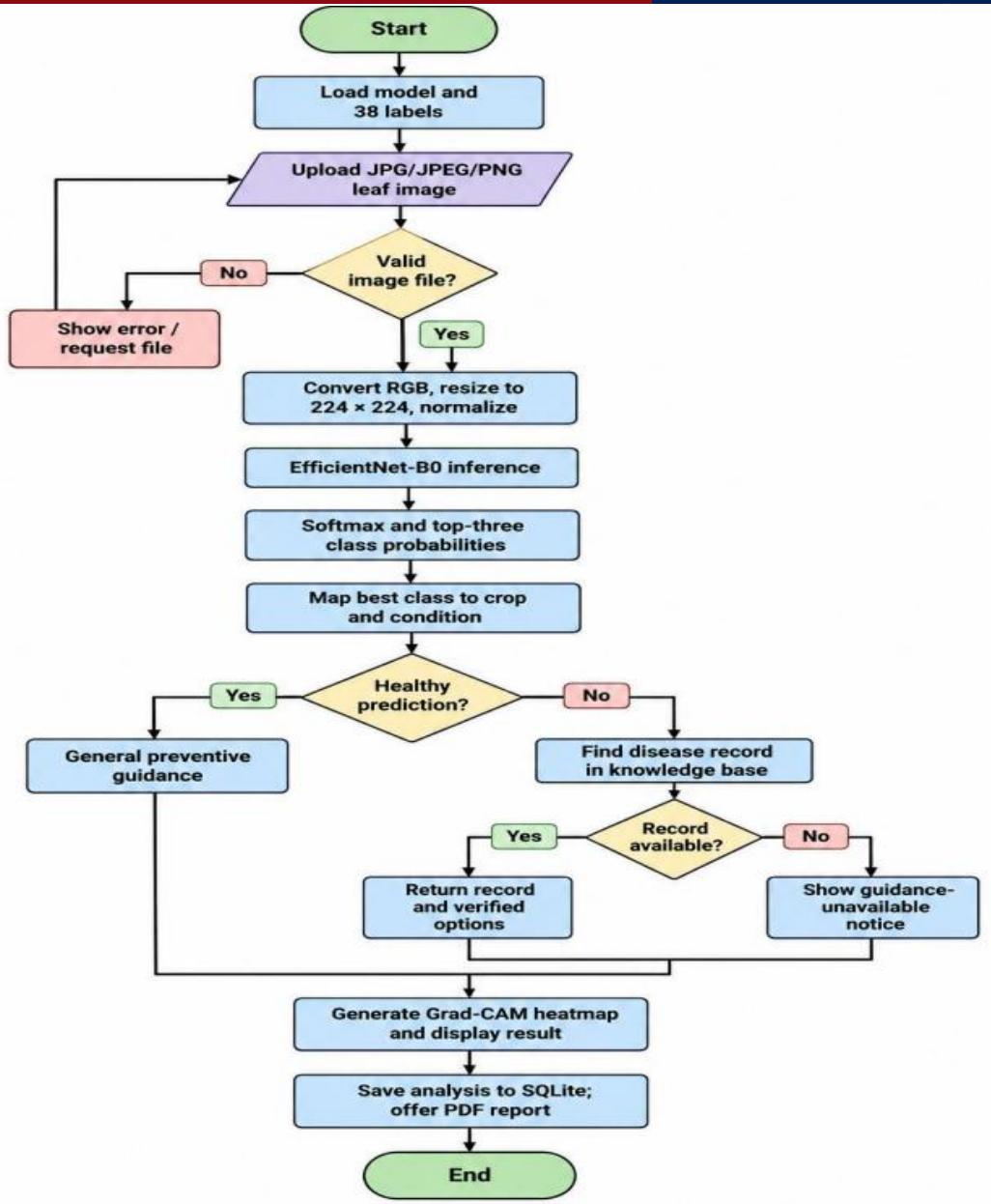
Hardware Requirements – Computer / Laptop,Minimum 8 GB RAM, CPU for model training

Software Requirements – Python, PyTorch, OpenCV, OpenCV, NumPy, Streamlit, Plotly, SQLite JSON and Pandas

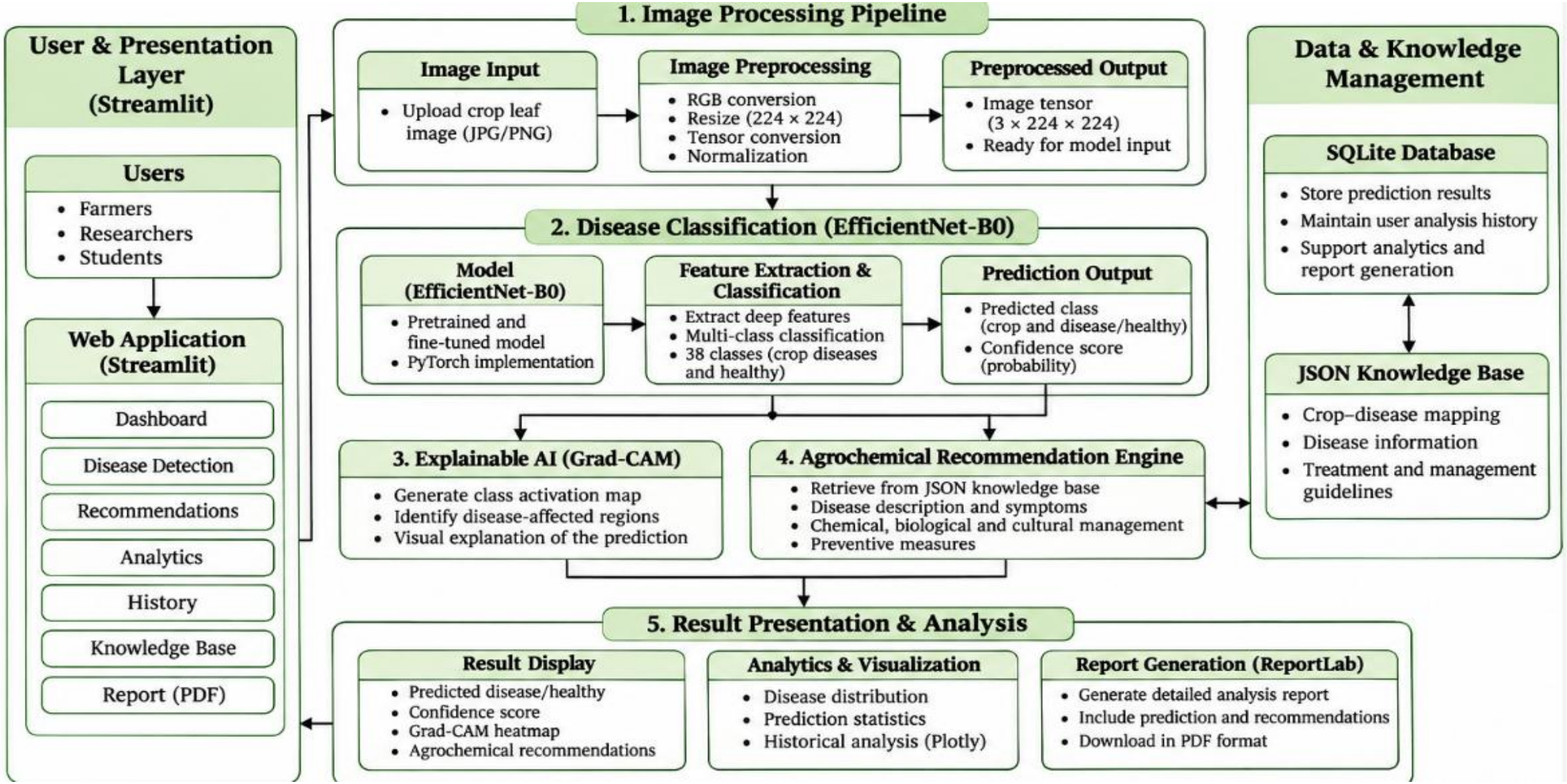
Dataset & Resource Requirements – PlantVillage crop leaf disease dataset,38 healthy and diseased crop classes, Labelled crop leaf images, Crop and disease information

Constraints – Limited coverage of crop and disease categories (38 classes), Variations in real-world leaf images and lighting conditions. Agrochemical recommendations require local guideline verification.

VIII. Flowchart



IX. System Architecture Architectural Diagram for the Proposed Framework



X. Proposed Methodology

Methodology Workflow

Step 1 - Image Input: Farmer uploads a crop leaf image through the application.

Step 2 - Image Preprocessing: Image is resized, normalized, and prepared for model input.

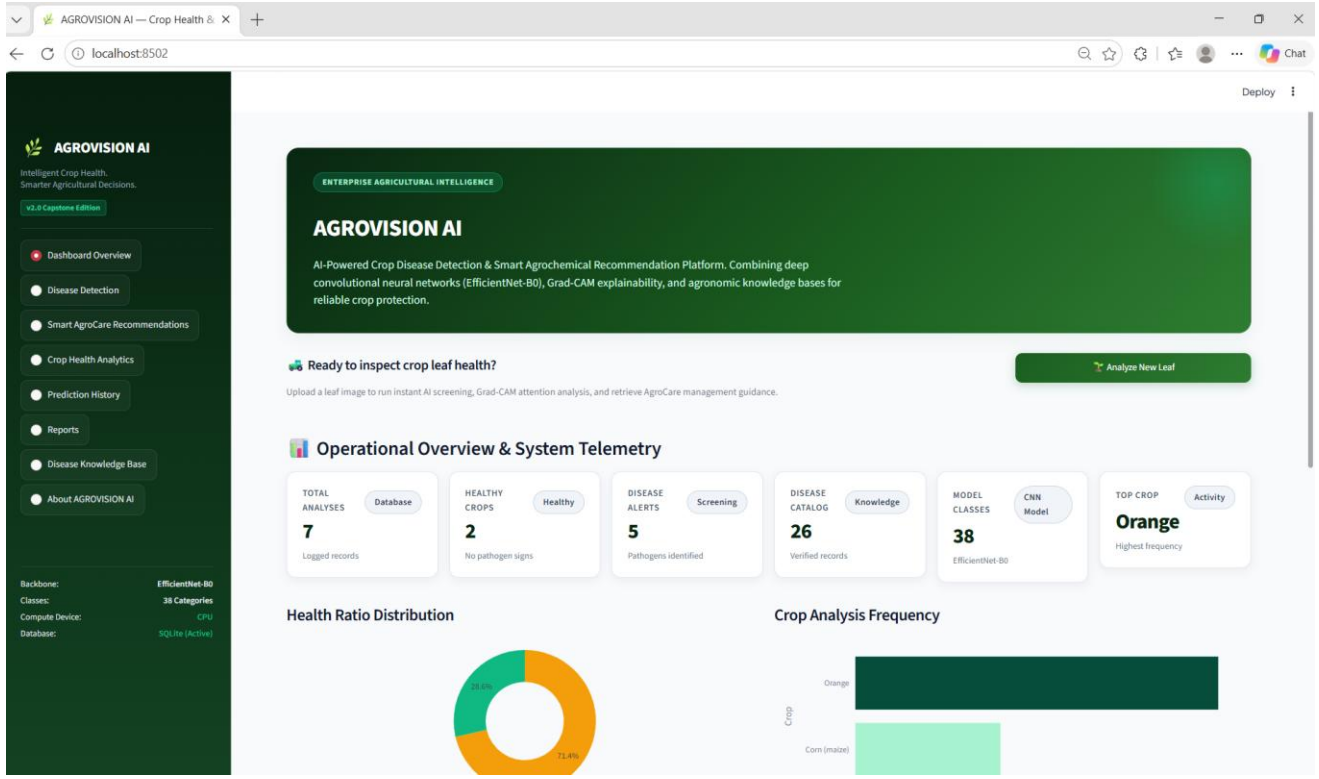
Step 3 - Disease Classification: The AI model identifies the crop disease or classifies the leaf as healthy.

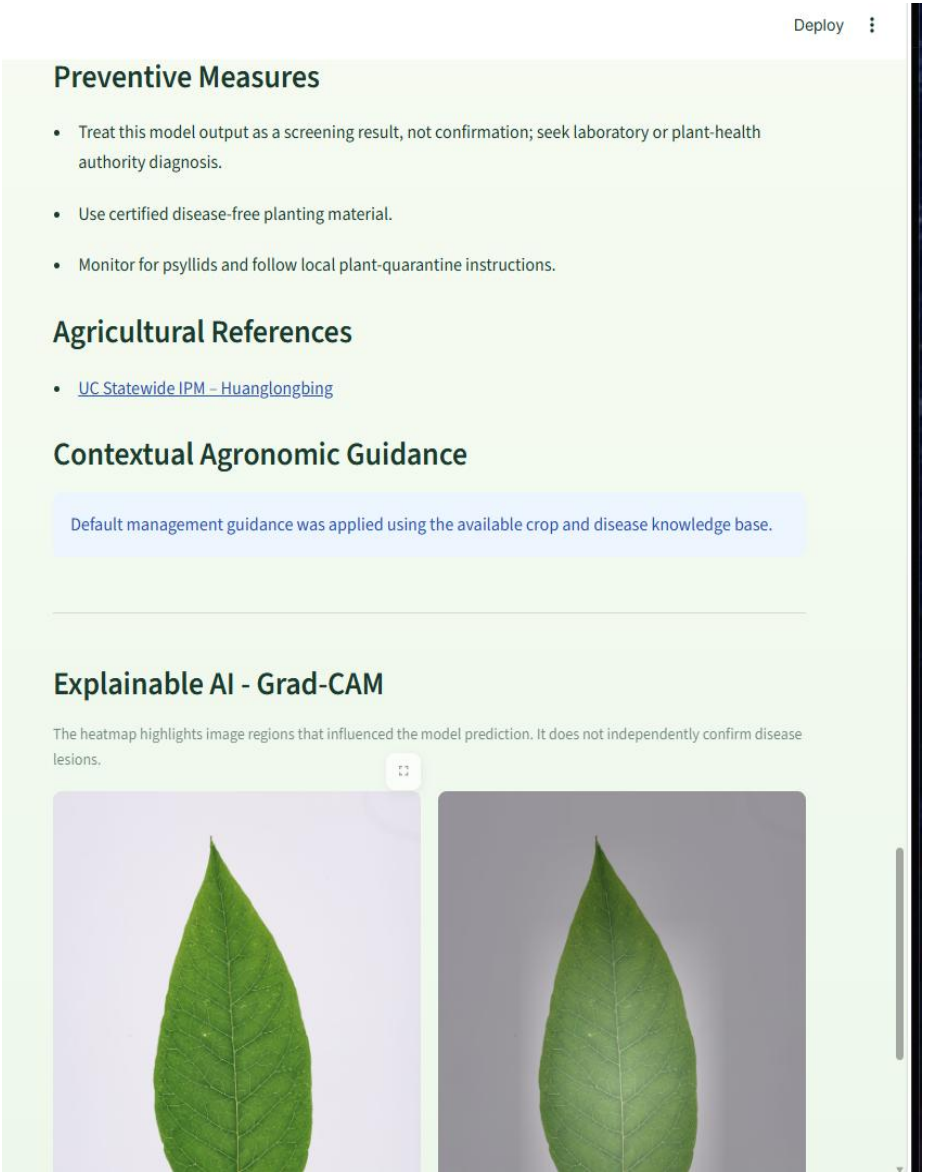
Step 4 - Agrochemical Recommendation: The system suggests suitable disease-management treatments based on the predicted disease and verified agricultural information.

Step 5 - Result Display: The application displays the predicted disease, confidence score, and relevant treatment recommendations.

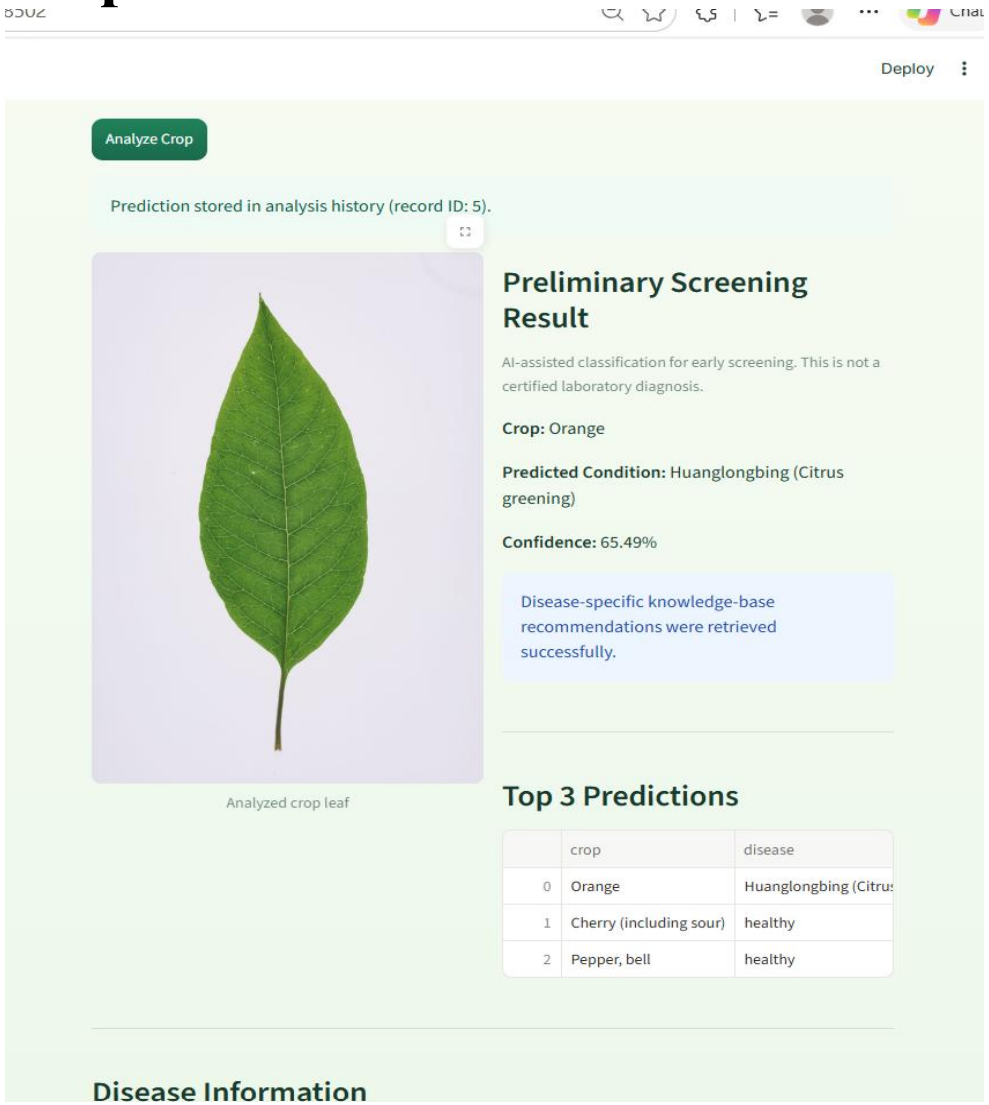
Output: An integrated AI-powered system for crop disease detection and smart agrochemical recommendation.

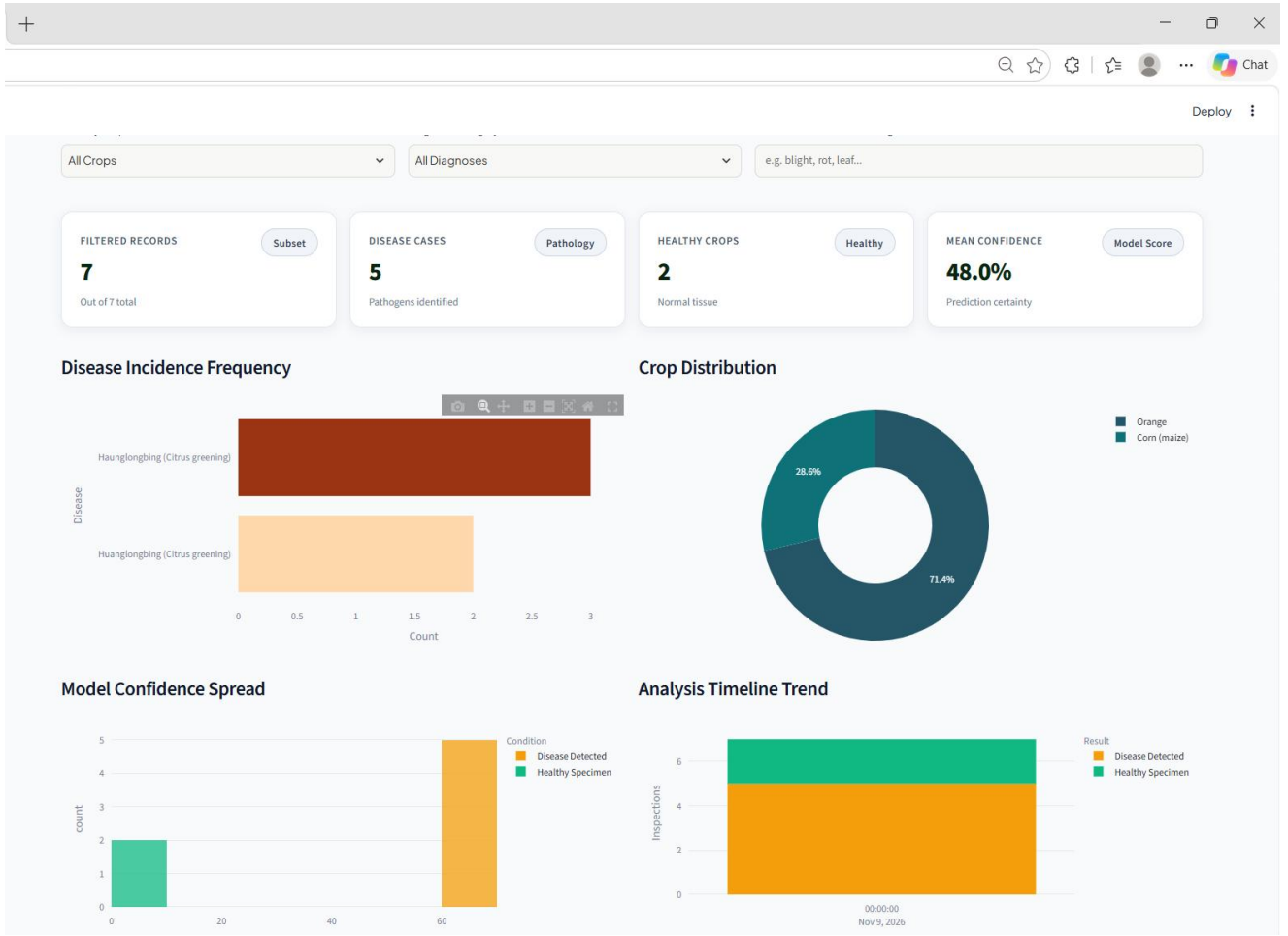
XI. Implementation





XI. Implementation





XII.Module Breakdown

Module 1: Dataset & Preprocessing
Collect and label crop leaf images.
Resize, normalize, and augment images.
Module 2: Disease Detection
Classify healthy and diseased leaves using deep learning.
Evaluate model performance using standard metrics.
Module 3: Agrochemical Recommendation
Map detected diseases to suitable treatments.
Use verified agricultural information for recommendations.
Module 4: Integration & Output
Integrate both modules into one application.
Display disease predictions and treatment recommendations.

XIII. Project work plan and timeline

Month	Module
Months 1–2	Dataset Collection, Preprocessing & Baseline Model
Months 3–4	CNN-Based Crop Disease Detection & Classification
Month 5	Agrochemical Recommendation Module
Month 6	Web/Mobile Interface Integration
Month 7	System Testing, Validation & Final Deployment

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THANK YOU

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